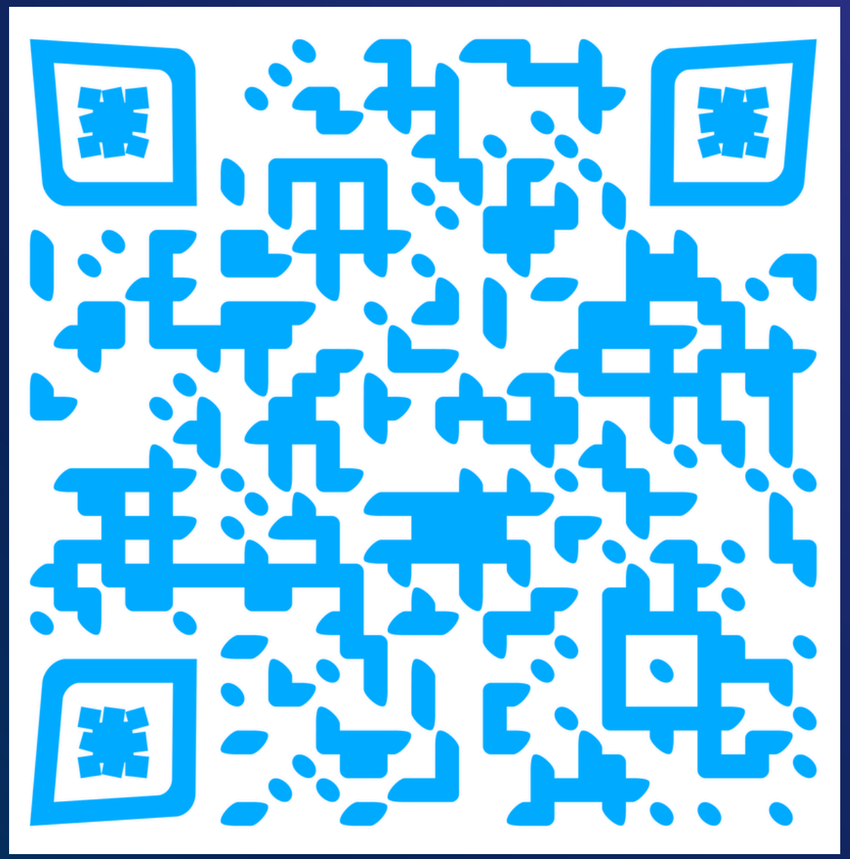
IRIS

INTEGRETED RESCUE AND INSPECTION SYSTEM



DRIVEN BY INNOVATION

IRIS

INTEGRETED RESCUE AND INSPECTION SYSTEM

DRIVEN BY INNOVATION

BY- SAI SWAROOP RAY

PROBLEM

In underground mines, slow and risky rescue operations put miners‘ lives in danger. Gas-related incidents and communication breakdowns further compound the dangers of the collapse of the mining caves. We urgently need a solution to enhance rescue efficiency.



Miners get trapped under the rocks for months. till the year 2026, no one has thought about their lives, and hence, more than 27,000 mining incidents have taken place, and more than several million miners have died.



EXAMPLE

Chinese Mining Accident during World War II (1942):
The disaster claimed the lives of an estimated 3,549 miners. The cause was insufficient food and medicines inadequate safety measures and lack of proper rescue operations.



Chilean Mining Rescue (2010):
A collapse trapped 33 miners approximately 2,300 feet (700 meters) underground.
This event gained global attention as the miners were trapped underground for a prolonged period of 69 days.



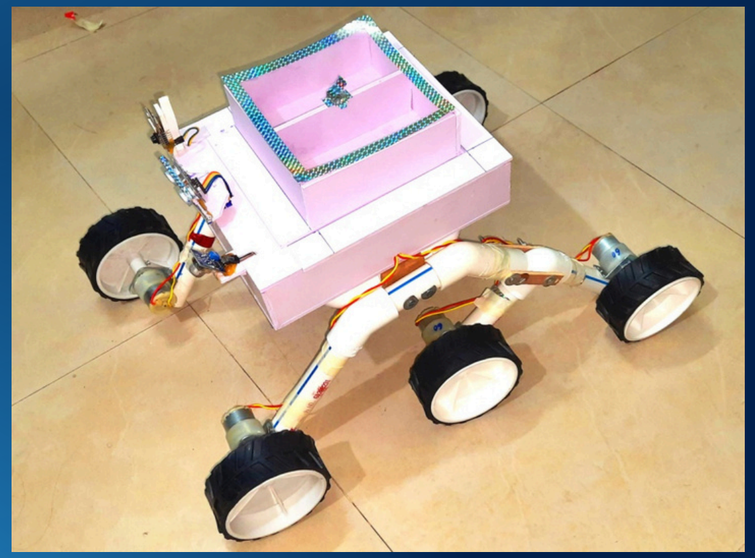
OBJECTIVE

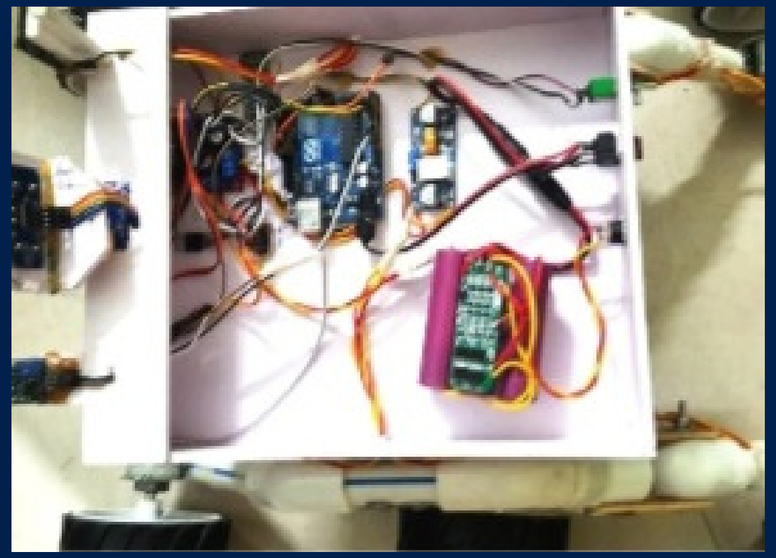
- Rapid Deployment: Quickly transported to disaster or mine sites.
- Environment Assessment: Cameras, LiDAR, and sensors detect hazards and obstacles.
- Terrain Navigation: 6-wheel rocker-bogie system and high-torque motors traverse rubble safely.
- Hazard Detection: Gas sensors identify dangerous gases and unstable structures.
- Live Monitoring: ESP32 Cam and mesh network provide real-time video and data.
- Operation Mode: Can operate autonomously or via remote Bluetooth control.
- Rescue Support: Delivers food, water, and medical aid to trapped miners.
- Data Reporting: Logs all data and footage to assist rescue planning.

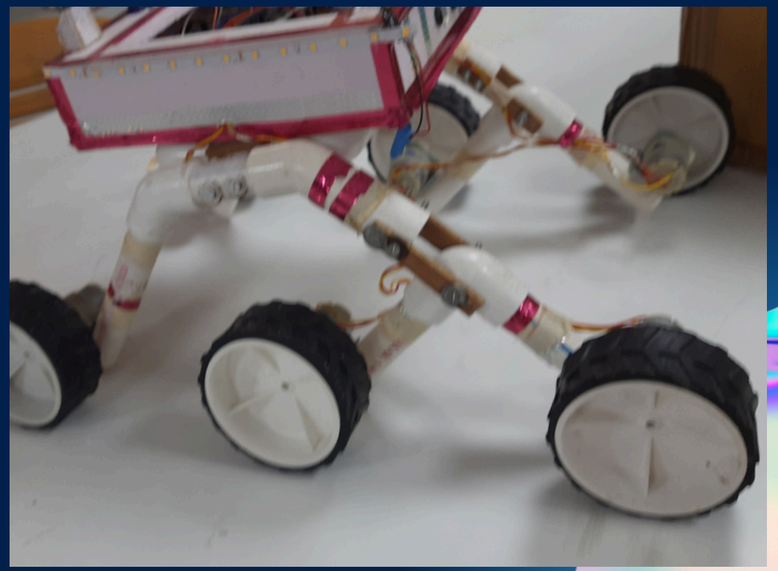
SOLUTION

IRIS

INTEGRETED RESCUE AND INSPECTION SYSTEM







INTRODUCTION

Introducing to IRIS:
INTEGRETED RESCUE AND INSPECTION SYSTEM

In the heart of the mining industry, where the depths of the Earth conceal both treasures and dangers, emerges a technological marvel known as "IRIS." Iris is a cutting-edge mining rescue robot designed to navigate the labyrinthine tunnels of mines and aid in critical rescue operations. With a range of advanced features and a mission-driven purpose, Iris stands as a beacon of hope in the most challenging of environments.

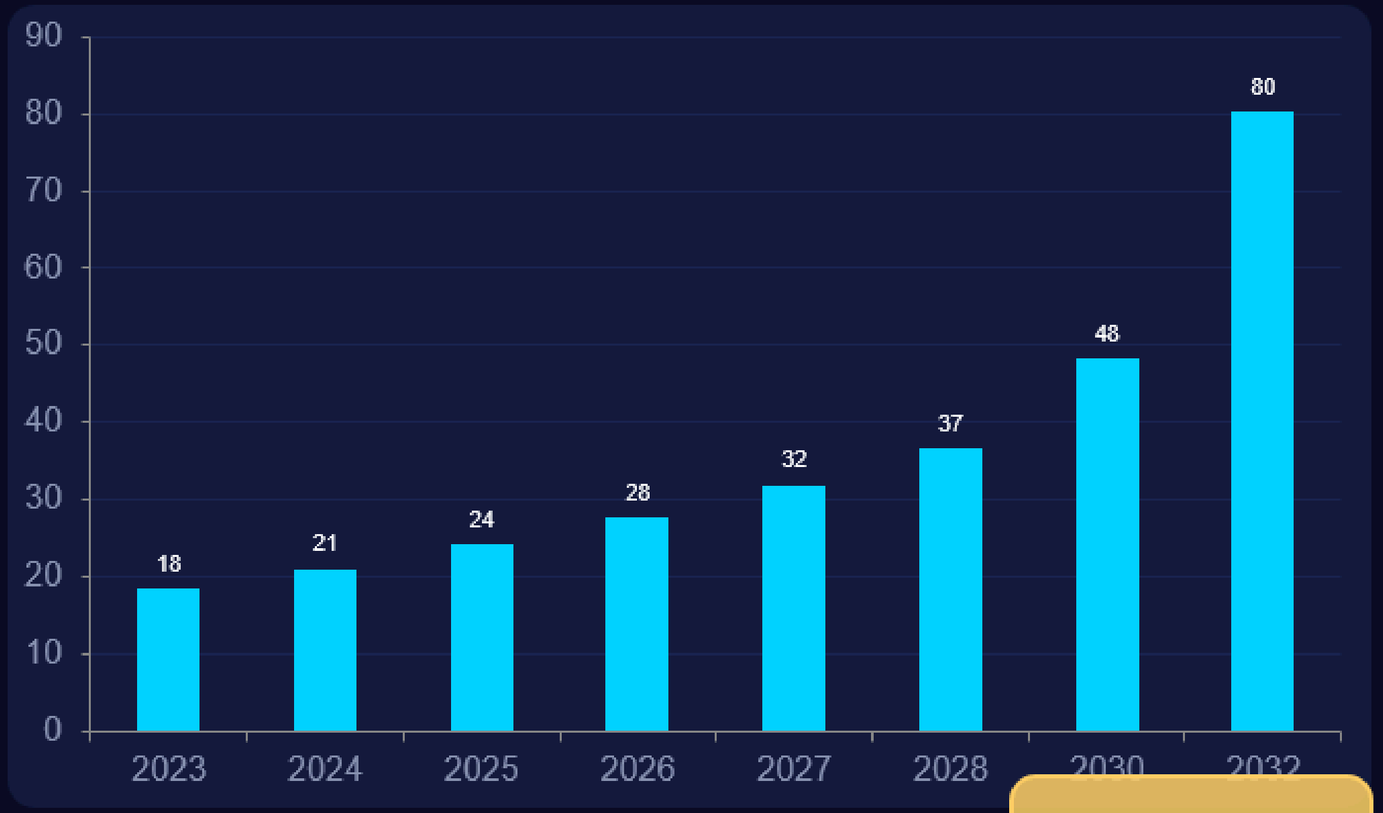
MARKET OPPORTUNITY

A $80 Billion Market Is Waiting.

The global search & rescue robotics market is experiencing explosive growth, driven by industrial safety regulations, increased mining activity, and government mandates for disaster preparedness.

TAM	**Total Addressable Market** Global rescue & industrial robotics by 2032	$80B+
SAM	**Serviceable Market** Underground mining safety & rescue systems	$18B
SOM	**Obtainable Market** India + SE Asia mining rescue modernization	$2B

MARKET GROWTH PROJECTION



14.79% CAGR

KEY GROWTH DRIVERS

- ✓ Rising mining accidents globally
- ✓ Govt. mandated safety robotics
- ✓ Industry 4.0 automation push
- ✓ Insurance risk reduction demand

COMPETITIVE ADVANTAGE

Why IRIS Wins Where Others Fail

Feature	IRIS	Manual Rescue	Conventional Drones	Industrial Cameras
Underground Navigation	✅ Full	⚠️ Risky	❌ None	❌ Static
Autonomous Operation	✅ Yes	❌ No	⚠️ Limited	❌ No
Gas Detection	✅ Real-time	⚠️ Manual	❌ None	❌ None
Supply Delivery	✅ Yes	⚠️ Slow	❌ No	❌ No
Mesh Communication	✅ Yes	❌ No	❌ No	❌ No
Cost Efficiency	✅ High	❌ Low	⚠️ Medium	⚠️ Medium
Human Risk	✅ Zero	❌ High	✅ Low	✅ Low

🎯 Only Robot Built FOR Mines

Not adapted from industrial bots — purpose-built for collapsed mine tunnels with the rocker-bogie system inspired by NASA rovers

🔗 Integrated System, Not Parts

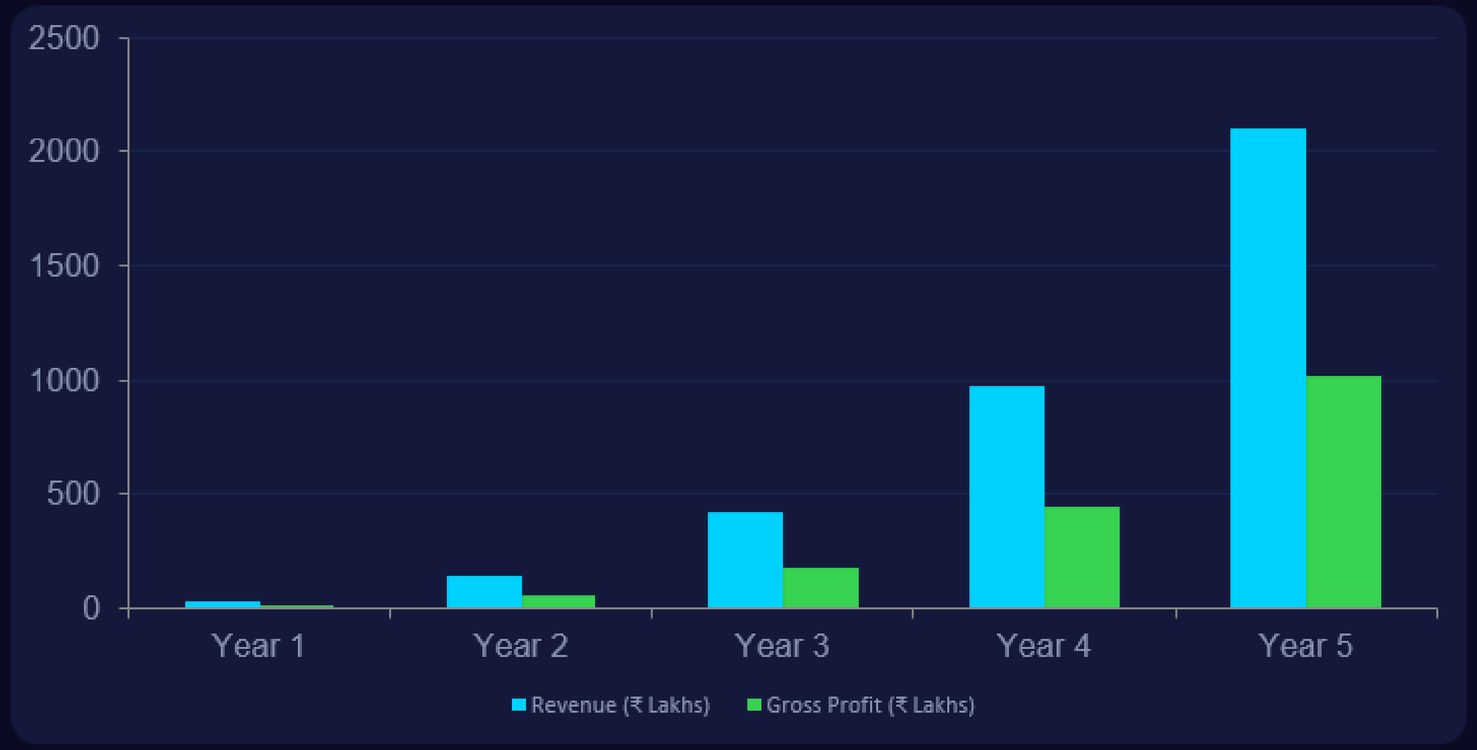
IRIS combines navigation, detection, communication, and delivery in one deployable unit — no coordination lag between systems

🌍 India-First, Scale Global

Priced for emerging market mining sectors (₹7L vs $200K+ western alternatives), yet built to international safety standards

FINANCIAL PROJECTIONS

Path to ₹10 Crore Revenue by Year 3



₹28L
Year 1 Revenue

₹4.2Cr
Year 3 Revenue

Month 18
Break-Even

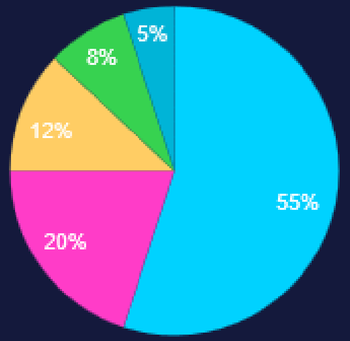
4 Units
Units Y1

60 Units
Units Y3

₹12.5L
LTV/Unit

KEY ASSUMPTIONS

- Y1: 4 units sold + 2 pilot programs (₹7L/unit + SaaS)
- Y2: 20 units + government tender + AMC renewals
- Y3: 60 units + India Coal/ONGC partnership + data analytics revenue
- Gross margin expanding from 35% → 49% as manufacturing scales



FUTURE ROADMAP

From Prototype to Global Rescue Standard

PHASE 1 · 0–6 Months

Validate & Refine

1. AI-powered autonomous navigation integration
2. Improved gas sensor suite (CO_2, O_2, H_2S)
3. First pilot with mining safety authority
4. IP filing for core innovations
5. Technical advisory board formation

PHASE 2 · 6–18 Months

Launch & Scale

1. Commercial production of 20+ units
2. Government tender submissions (Coal India, NDMA)
3. SaaS dashboard launch with analytics
4. Series A funding round
5. Team expansion (10 members)

PHASE 3 · 18–36 Months

Dominate Market

1. Global expansion to SE Asia, Africa mining markets
2. IRIS v2: aerial + ground hybrid system
3. Predictive hazard AI trained on mine data
4. Strategic partnerships with Komatsu, Caterpillar
5. Target 500+ units deployed globally

WORKING

- Bluetooth HC-05 Module: Enables wireless communication for remote control, data transfer, and monitoring sensor readings.
- Automatic Mode: Allows IRIS to operate autonomously using sensors and programming to navigate and perform tasks.
- Gas Detection Sensor (MQ-2): Detects hazardous gases like methane, carbon monoxide, and propane to ensure miner safety.
- 6 Wheels: Provide stability and traction for navigating rugged terrain.
- ESP32 Cam: Streams real-time video for surveillance and rescue decision-making.
- LiDAR Sensor: Maps surroundings and detects obstacles for precise navigation.
- Food & Medical Delivery System: Allows IRIS to provide water, food, or medical supplies to trapped miners.
- Mesh Network System: Ensures live video and sensor data transmission even in deep mines where signals are weak

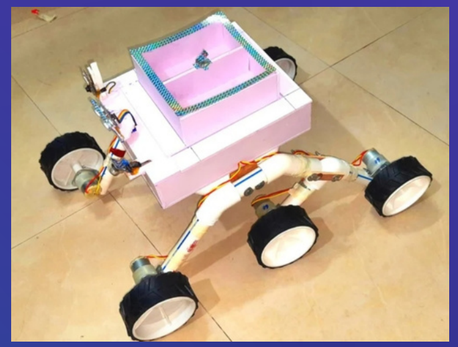
MATERIALS USED

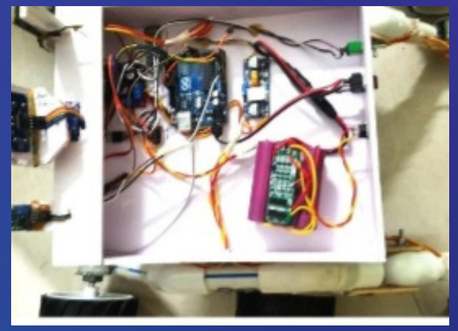
- Arduino Uno
- L298N Motor Driver
- Ultrasonic Sensor
- MQ-2 Gas Sensor
- ESP32 Cam (ESP32CA)
- PVC (Polyvinyl Chloride)
- Wheels
- Lithium-ion Batteries
- Chassis
- Bluetooth HC-05 Module
- Other Electronic Components

COST ANALYSIS

PROTOTYPE

2,365





REAL MODEL

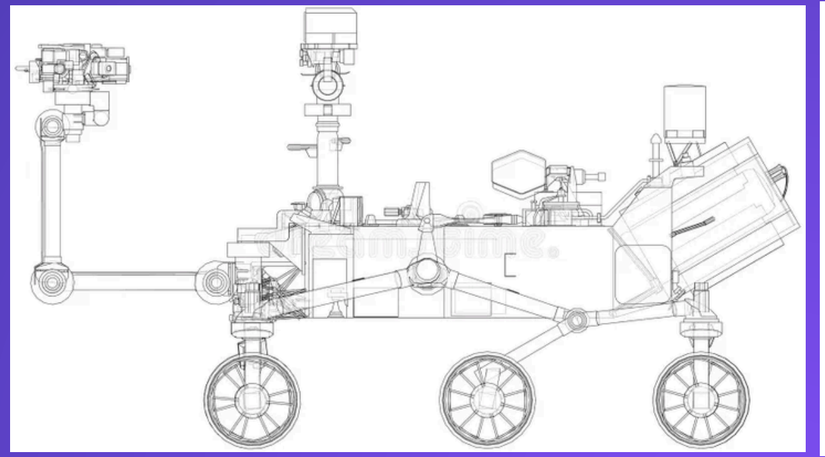
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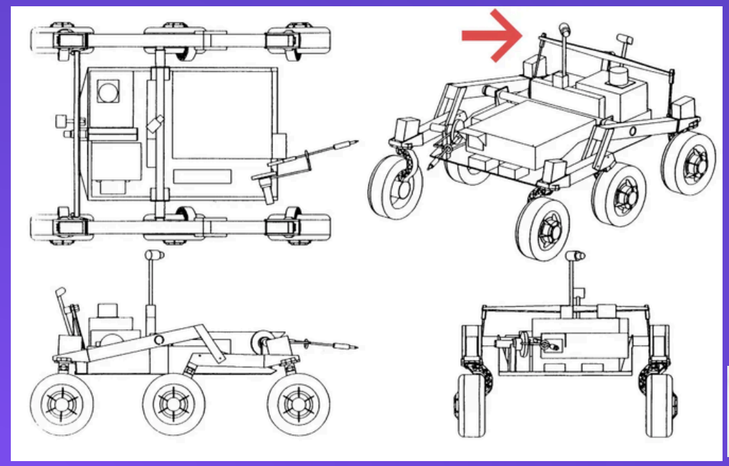






DESIGN OF IRIS







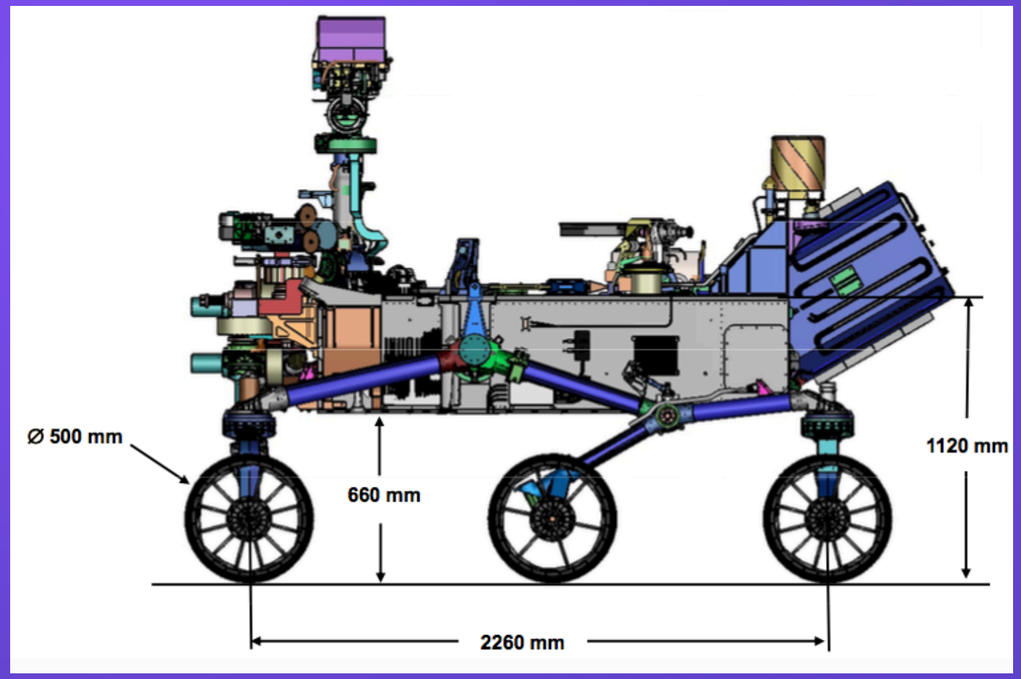














BUISNESS & REVENUE MODEL

IRIS is a state-of-the-art rescue robot designed to navigate extreme terrains, detect hazards in real-time, and provide rapid assistance in mines, disaster zones, and collapsed structures. Its advanced sensors, high-torque mobility system, and secure mesh network for communication make it faster and safer than conventional methods. The prototype costs ₹4.5 lakh to build and can be sold for ₹7 lakh per unit, offering a profitable margin. With the global rescue robot market projected to grow from USD 29.6 billion in 2025 to over USD 80 billion by 2032, IRIS has strong revenue potential through direct sales, subscription services for software updates and data analytics, training programs, maintenance contracts, and strategic partnerships with disaster response agencies.



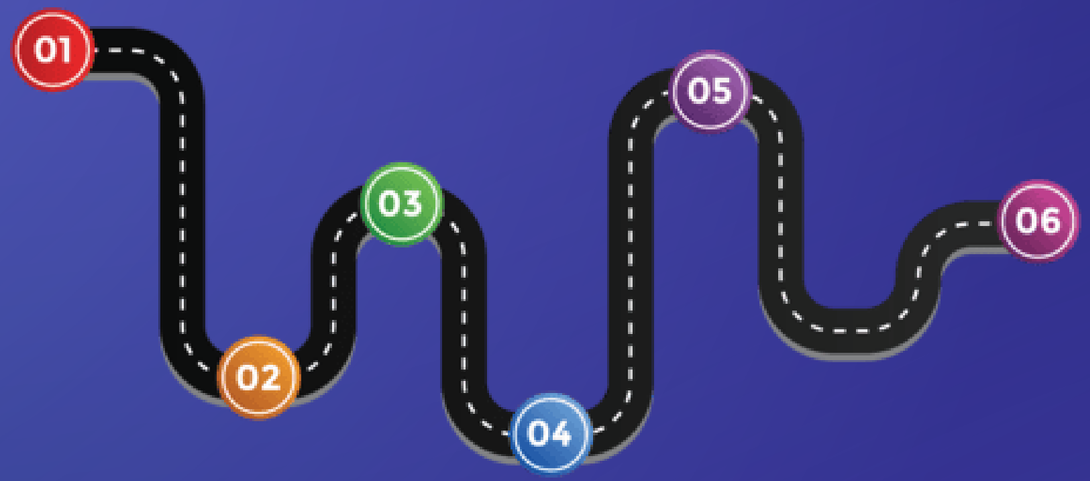
Components	Estimated Cost /-(INR)
Custom PCB & Processing unit	1,00,000
LiDAR & Navigation System	50,000
AI-Powered Camera System	25,000
Gas & Environmental Sensor	5,000
High-Torque Motors & Controllers	60,000
Long-Lasting Battery & Power	1,00,000
Communication Modules	10,000
Chassis (Carbon-Fiber and metal)	55,000
Assembly & testing cost	10,000
Total	4,50,000

UNIQUE SELLING POINT

IRIS is the only rescue robot designed for extreme environments, combining rugged terrain navigation, real-time hazard detection, and a secure mesh network for instant communication with rescuers. Its advanced sensors, high-torque mobility system, and modular design allow it to reach trapped individuals in mines, disaster zones, and collapsed structures faster and safer than any conventional rescue method.

FUTURE ROADMAP

Enhance sensors for gas, temperature, and motion detection. Integrate AI for autonomous navigation and hazard analysis. Add advanced communication modules for faster mesh network updates. Introduce modular attachments for lifting, carrying, or delivering supplies. Develop software for remote monitoring, data analytics, and predictive rescue planning.





CONCLUSION

IRIS is highly effective in mining rescue operations due to its advanced capabilities. Its rocker bogie system allows smooth navigation across uneven underground terrain, while the MQ-2 gas sensor detects hazardous gases to ensure safety. It can deliver essential supplies like food, water, and medicine to trapped miners, and the ESP32 camera provides real-time video surveillance for informed decision-making. Equipped with ultrasonic sensors, IRIS also operates in autonomous mode, reducing the need for constant human control and making rescue missions faster and safer.




Thank you